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Supporting organizational decisions on How to improve customer repurchase using multi-instance counterfactual explanations

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ABSTRACT

Improving customer repurchase intention constitutes a key activity for maintaining sustainable business performance. Returning customers provide many economic and other benefits to businesses. In contrast, attracting new customers is a process that is associated with high costs. This work proposes a novel counterfactual explanations methodology that utilizes textual data from electronic word of mouth to recommend business changes that can improve customers' repurchase behavior. Counterfactual explanation methods gained considerable attention because their logic aligns with human reasoning and the fact that they can recommend low-cost actions on how to turn an unfavorable outcome into a favorable. Most counterfactual explanation methods however recommend actions that can change the outcome of individual instances (i.e. one customer) rather than a group of instances. Therefore, this work proposes a multi-instance counterfactual explanation method that recommends optimum changes to an organization's practices/policies that increase repurchase intention for many customers or customer segments.

The proposed methodology utilizes topic modeling to extract customer opinions from online reviews' text and use topics as features to train a binary classifier that predicts customer revisit intention. Multi-instance counterfactual explanations are computed for all or different groups of non-revisiting customers, recommending optimum business changes that can increase revisit intention. The proposed methodology is empirically evaluated through a case study on the restaurant revisit problem and compared against a prominent alternative from the literature. The results show that the method has better performance to the alternative method and produces recommendations that are actionable and abide by the customer-repurchase literature.

1. Introduction

Machine learning (ML) and Artificial Intelligence (AI) have been applied to many different business problems aiming to improve organizational competitiveness through improved performance. A critical issue in AI and its applications in business and other domains is the lack of interpretability of highly complex models such as ensemble machine learning and deep learning models that are usually used. These models have good predictive performance but their complex structure hinders decision makers' understanding of how they make predictions. Because of this, they are referred to as “black boxes”, since it is humanly impossible to understand their internal structure and how they reason without using eXplainable AI (XAI) techniques [1]. XAI assists in interpreting and understanding such complex models and thus contributes to their validation (free from biases), which in turn increases decision-makers' confidence and trust in their predictions [1]. XAI is thus important during model development by assisting in the extraction of patterns from complex models' logic to improve performance and prevent overfitting. In this work, we propose and apply a novel XAI methodology to the business problem of customer repurchase, to identify optimum changes to business practices that will improve customers' repurchase intention. In this work, we utilize electronic word of mouth (eWOM), such as online reviews, shown in Fig. 1, since it is a known factor that affects customer purchase decisions/intentions and products/services development/improvement [2]. The proposed counterfactual explanation method analyses eWOM and produces recommendations that are easily understood by decision-makers and thus assist in optimizing business change [3] that is necessary for

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businesses to remain competitive.

Various XAI approaches [3–6] have been proposed for explaining models' outputs that vary in scope(local, global), flexibility (model-specific, model-agnostic), and output type (numeric, categorical). One category of methods are the model-based techniques, where the analyst intentionally selects an interpretable model such as a small decision tree or a linear model. The second category is based on the post-hoc methods, which assume that a trained model exists and post-training analysis is performed to infer the model's internal logic using different inputs and observed outputs. A popular technique of this category are counterfactual explanations [7]. These mimic human explanations and reasoning [8] and can be interpreted as minimal actionable recommendations (small and easily realized) that change the model's outcome to some desired state. Thus when applied to business problems such as customer repurchase, their explanations recommend optimum modifications to business practices to maximize goal attainment (in our case increase repurchase intention) with minimum cost. An important limitation of counterfactual explanations however is that they usually fail to find the optimum action that will satisfy multiple entities (multi-instance) [9], such as customers, suppliers, etc. Instead, they can only find the optimum change for single cases (one customer). Thus, a new line of research on multi-instance counterfactual explanations methodologies emerged [10–12] and is gaining considerable attention, that addresses this problem. Existing multi-instance methods however have limitations, such as bad performance [10] or limited support to data from different domains [11,12]. For instance, the approach described in [12] does not support binary data sets.

1.1. Contributions

In this work, we address the problem of customer repurchase by proposing a methodology that extracts customers' opinions from electronic word of mouth (eWOM) and recommends changes to organizations' practices that will increase customers' repurchase intention.

The paper introduces a new concept, called *set consistent counterfactual explanations* to address the multi-instance counterfactual explanations problem. In our proposed methodology we mine customers' opinions from eWOM (i.e. textual reviews), using topic modeling, and use the proposed concept to generate recommendations for businesses on how to increase customers' repurchase intention. The repurchase problem is known to have important effects on business sustainability and profitability since revisitors require less marketing costs to be convinced to make a purchase compared to new customers. To the best of our knowledge, no previous work addresses this business problem through counterfactual explanations. Moreover, despite knowledge of the various individual factors that influence revisit in the literature [13,14], no previous work recommends optimum changes of such factors to increase customers' revisits, with most previous studies utilizing questionnaires, seeking to identify or confirm antecedents variables of revisit, such as the work in [15,16].

The remainder of this work is structured as follows: After reviewing the literature on customer repurchase and explainable AI in Section 2, we briefly discuss the foundations of counterfactual explanations and topic modeling in Section 3. The formalization of the concepts used in the proposed method are presented in Section 4. Section 5 introduces the proposed methodology for generating recommendations on how to improve customer repurchase. An empirical evaluation of the method, through a case study, is presented in Section 6. All mathematical proofs and derivations are included in Appendix A.

2. Literature review

This section describes the literature with regards to customer repurchase and XAI.

2.1. Customers repurchase

Customer repurchase is a key business issue that affects sustainability and profitability [17] and constitutes a critical success factor for many types of businesses. For instance, in the context of the service industry (e.g. restaurants), the effect of repurchase is significant, since the marketing costs of acquiring new customers and bringing them to the level of profitability of existing customers (repeating customers), is sixteen times greater than that of keeping existing customers [17]. Evidence from literature [17] also shows that repeating customers spend more, have more realistic expectations, are satisfied more easily, tolerate service errors, and are more likely to spread positive electronic word of mouth (eWOM) [18], as shown in Fig. 1.

Customers' decision to repurchase is formed after comparing the value gained from a repurchase against the cost and risks of attaining a different product or service. This is influenced by different aspects such as the loss aversion property, which asserts that greater weight is given to losses than gains [19], thus loss significantly influences repurchase behavior. Central to the concept of repurchase (revisit in the context of venues) is the notion of satisfaction [17], which is affected by the difference between the perception after experiencing a service, and the expectation prior to the experience [20]. Satisfaction is also linked to value gain, with more satisfied customers being more likely to revisit and spread positive eWOM. Due to the link between satisfaction and revisit, most previous studies use satisfaction as a proxy to revisit intention and utilize its antecedents such as perceived value, service quality, image of the place, authenticity, memorable experience, trust, risk, etc., to assess it [21].

Prevalent theories applied to the analysis of customer repurchase intention, include the Theory of Planned Behavior (TPB) [22] and the three-factor theory (TFT) [23]. These however, mainly use

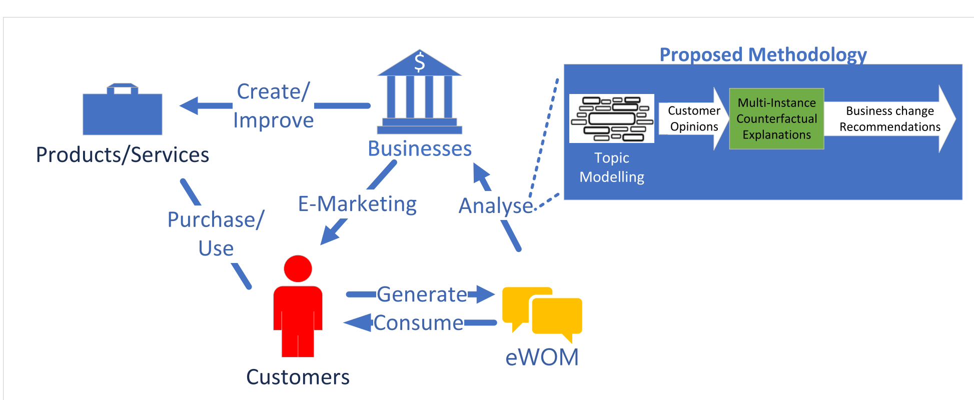


Fig. 1. Illustration of the problem and our proposed methodology.

questionnaires to analyze repurchase intention [22], with the TFT examining the asymmetric impact [24] of service/product attributes on customer satisfaction by considering certain venue properties as expected and others as value-adding [23]. Thus, low performance on an expected attribute may create a stronger negative influence on satisfaction than on a value-adding attribute. For instance, customers will be dissatisfied if a restaurant is dirty, but most likely they will not be satisfied if the restaurant is clean, since this is expected. Similarly, if a restaurant is clean but the staff does not make customers feel special, the negative effect of the latter is not as significant as the effect of low cleanliness. The TFT groups satisfaction into three categories of factors: basic, performance, and excitement. Basic factors refer to the minimum requirements that cause dissatisfaction if not fulfilled but do not lead to satisfaction if fulfilled or exceeded. Performance factors lead to satisfaction if performance is high and to dissatisfaction if performance is low. Excitement factors increase customer satisfaction when present, but do not cause dissatisfaction when absent. Such attributes need to be identified early when analyzing customers' repurchase intention and addressed in a fashion so that basic factors are satisfied first, before addressing performance and excitement factors.

Factors that affect satisfaction have also been described in terms of the physical environment (of the venue), the offered service [25], the staff, and other customer attitude, all of which compose the servicescape of a venue [26]. Physical servicescape refers to factors such as the venue's aesthetics/design, layout/seating, comfort, and ambient conditions (music, temperature, comfort, etc.) and has been found to influence customers' experience and repurchase intention. The social dimension of service is categorized into employee servicescape and customer servicescape [14]. The latter refers to interactions between other customers and the former to interactions with employees. These have been found to influence customers' emotions, satisfaction, perception of service and revisit intention [26]. Customer servicescape is based on social identity theory portraying that people only desire to engage with others they consider as similar to themselves. The perceived similarity is defined as “the degree to which a customer feels that he is similar to other customers”. This leads to customers feeling comfortable/compatible when surrounded by customers with similar features to themselves. The literature also highlights that psychological motives prevail over physical attributes of service delivery when it comes to revisits. Such servicescape information can be obtained from electronic word of mouth (eWOM) through techniques such as topic modeling, as described in Section 3.2. EWOM constitutes a key source of data that businesses can harness to infer customers' thoughts & opinions, and understand the factors that lead to repurchase. EWOM lends itself to automation and thus provides an expedited approach to analyzing servicescape from customers' perceptions expressed in textual form [13], and has many advantages over traditional techniques such as questionnaires. We argue that the combination of customers' opinions extracted from eWOM with counterfactual explanations can assist businesses in identifying the optimum changes they need to make, in terms of servicescape, to improve customer repurchase [27].

2.2. Explainable AI and counterfactual explanations

EXplainable AI (XAI) [1,28] methods aim to make AI-based systems transparent by explaining the reasoning behind the behavior/output of the system to the user. However, it is still not fully understood what exactly makes up a good explanation [29], therefore, the XAI community has developed a wide variety of different explanation methods over the last few years [28]. Consequently, the explanation method to be used in different cases must be chosen carefully.

Besides interpretable-by-design models, such as low-complexity decision trees or low dimensional linear models) [28], a prominent class of explanation methods that provide recommendations based on models' output are the post-hoc approaches. These methods are usually model-agnostic, meaning that they do not need to examine the inner workings of a model's logic, and are therefore applicable to a wide range of ML models [28]. Another category of explanation methods are local and global methods [28]. Global methods try to explain the entire system, while local only explain a specific decision or how the system behaves in a small region in the data space. Most post-hoc methods are local methods that extract local explanations from complex models [28]. Prominent instances of such post-hoc explanations that are characterized as local/global or both are, LIME [30], SHAP [31], and counterfactual explanations [7].

2.2.1. Counterfactual explanations

Counterfactual explanations [7] have attracted significant attention in recent years [9,32–35] because they mimic human explanations [8] and therefore are easy to understand, and are also able to identify optimum changes that can lead to desired (i.e. favorable) outcomes. A counterfactual explanation [7] provides recommendations on how to alter the prediction of a system to the desired outcome – e.g. how to turn a rejected loan application into an accepted application. However, “plain” counterfactual explanations do not address problems in which we have to assign actions to satisfy multiple instances (e.g. customers) simultaneously [10,12,36,37], nor to the case where several decisions must be explained at the same time [38]. In such cases, an approach is required that can generate a counterfactual explanation that assigns actions to multiple instances in a transparent and consistent way [10,12]. This is the problem we address in this paper.

2.2.2. Multi-instance counterfactual explanations

Multi-instance counterfactual explanations refer to the generation of counterfactual explanations that apply to multiple instances. They aim to provide explanations for different groups of instances (e.g. categories of customers) – i.e. one counterfactual that is valid for a group of instances. Such an approach has many practical implications for different business problems. For instance, in the context of customers' repurchases, groups of customers can have different opinions about the servicescape of a particular restaurant/destination/chain of restaurants, with one group complaining about the speed of service and another about the attitude of personnel. To satisfy the group of customers that the business considers as the most important for its survival, it can utilize multi-instance counterfactual explanations to generate actionable recommendations, which will then satisfy all customers in that group.

In contrast to counterfactual explanations, multi-instance counterfactuals are a novel concept and consequently, existing work on this is limited. For instance, [39] apply multi-instance counterfactuals to employee attrition but only consider a linear classifier. Work by [12] proposes a counterfactual explanation tree, which assigns counterfactual explanations to a learned decision tree, which assigns samples to groups (i.e. each learned leaf in the tree is interpreted as a group). This method is only applicable if an automatic grouping into sub-groups is required but the user has no control of the grouping. Furthermore, binary data domains are not supported. The authors of [36] propose to generate a set of basic explanations that are afterward used to construct explanations between groups. In general, a large part of existing work for multi-instance counterfactuals can be interpreted as summarizing or aggregating local counterfactual explanations [10,37,40]. In [10], multi-instance counterfactuals are generated by first computing individual counterfactuals and then selecting one that maximizes satisfaction for a given set of samples, for which a multi-instance counterfactual is requested. Similarly to [10], the authors of [37] compute global counterfactuals by introducing instance-specific scaling for a set of counterfactual explanations, such that, most of the instances are satisfied.

3. Foundations

This section provides background knowledge regarding counterfactual explanations and topic modeling, which are part of our proposed methodology that is described in Section 5.

3.1. Counterfactual explanations

The formalization and computation of counterfactual explanations involves the consideration of two important aspects [7]: 1) the contrasting property, that the output of the system indeed changes, and 2) the cost of the counterfactual – i.e. the cost and effort it takes to execute the explanation (i.e. recommendations) in the real world. Both properties have been combined (in a non-causal setting) [7] into a single objective optimization problem as stated in Definition 1.

Definition 1. ((Closest) Counterfactual Explanation [7]). *For a prediction function* $h : \mathbb{R}^d \to \mathscr{Y}$ *and an instance* $\vec{x}_{orig} \in \mathbb{R}^d$*, the closest counterfactual explanation states the smallest change* $\vec{\delta}_{cf} \in \mathbb{R}^d$ *to* $\vec{x}_{orig}$ *that changes the original output* $h\left(\vec{x}_{orig}\right)$ *to* $h\left(\vec{x}_{orig} + \vec{\delta}_{cf}\right) = y_{cf} \neq h\left(\vec{x}_{orig}\right)$:

$$\underset{\vec{\delta}_{cf} \in \mathbb{R}^d}{argmin}\ \ell\left(h\left(\vec{x}_{orig} + \vec{\delta}_{cf}\right), y_{cf}\right) + C \cdot \theta\left(\vec{\delta}_{cf}\right) \tag{1}$$

where $\ell(\cdot)$ *denotes a loss function that penalizes deviation from the requested output* y_{cf}, $\theta(\cdot)$ *implements the cost of* $\vec{\delta}_{cf}$*, and* $C > 0$ *denotes the regularization strength to merge the two objectives (change in prediction and small change) into a single cost function.* Note that the cost of the counterfactual, here modeled by $\theta(\cdot)$, is highly domain and use-case specific and therefore must be chosen carefully in practice and might require domain knowledge. In many implementations & toolboxes [32], the *p*-norm is used as a default.

Note that a counterfactual explanation provides precise recommendations regarding the magnitude of change required to be applied to an instance to obtain the desired output. This is an advantage over linear models that are considered interpretable but the degree of change can not be inferred merely from the feature weights.

Besides those two essential properties (contrasting and cost), there exist additional relevant aspects such as plausibility [35,41], diversity [33], robustness [42], fairness [43], etc. which have been addressed in literature [32]. However, the basic formalization Definition 1 is still very popular and widely used in practice [32,44].

Numerous toolboxes exist for computing counterfactual explanations [32] – most include some additional aspects such as plausibility [35,41], diversity [33], etc. However, most of them focus on providing a single recommendation or multiple diverse recommendations [33] for a given single instance only (see Definition 1) – but they do not address the case where we have to assign actions to multiple instances simultaneously [12]. This problem is not specific to counterfactual explanation methods but generalizes to local explanation methods in general: One of the critical problems is that the region where the explanations can and should be applied is unclear [45,46]. Due to this issue, local explanation methods often fail to explain the global behavior of complex models. To address the above issue, a few approaches for a global summary of local explanations have been proposed [47]. Currently, one can observe the first attempts of computing counterfactual explanations that are valid for a group of instances [10,12,37,39].

3.2. Topic modeling

Topic modeling [48] refers to unsupervised machine learning techniques that can identify opinions (themes) in a collection of documents, such as reviews, by finding sets of words that frequently occur together. Topic modeling constitutes a popular tool for extracting information from unstructured data such as electronic word of mouth (eWOM), and in this work, we use it to identify customers' opinions from online reviews and how these relate to servicescape [14]. The two main groups of topic modeling techniques are probabilistic and neural-network-based. The former are Bayesian probabilistic models that assume that the distributional properties of words in documents dictate their semantic meaning (distributional hypothesis). In such models, a word is characterized by its surrounding words. During topic model learning words are assigned to topics (similar topics use similar words) and topics to documents(documents have several topics). The neural-topic models use optimization to infer topics and are more flexible and scalable than probabilistic but are less interpretable. Thus in this work, we use a mainstream probabilistic technique, namely, Latent Dirichlet Allocation (LDA) [49].

The process for learning a probabilistic topic model initiates with the removal of common and custom stop-words and irrelevant information such as punctuation. The process continues with the breaking of sentences into word tokens (tokenization), and converting words to their root form (stemming), which can be omitted if it does not contribute to the topics' interpretability. To identify the optimum number of topics that best fits the corpus an iterative process is followed examining the performance of the model using different numbers of topics. The evaluation metric that is usually employed for topics to be interpretable and meaningful to humans is semantic coherence, which measures the degree of semantic similarity between high-scoring words in a topic [50].

4. Multi-instance counterfactual explanations

In this section, we introduce our *set consistent counterfactual explanations* for modeling multi-instance counterfactual explanations, which cover (i.e. explain) more than a single instance. We provide a high-level overview of this in the following sub-sections and refer the interested reader to Appendix A for more (mathematical) details.

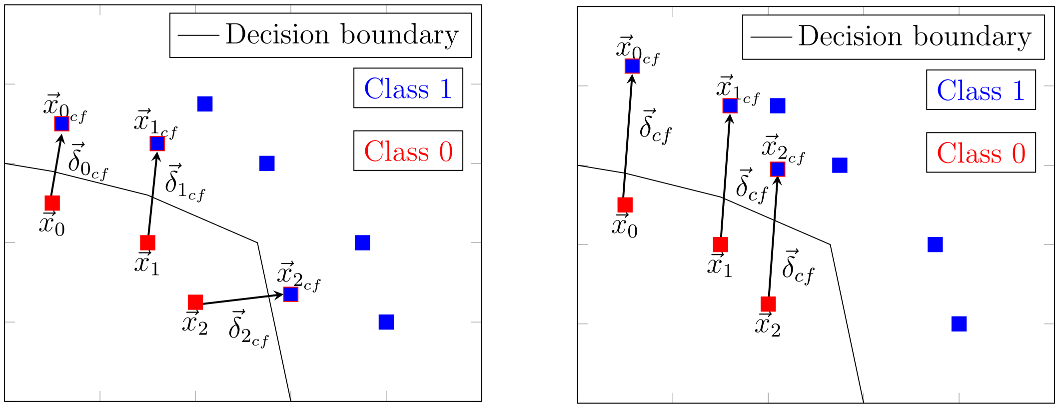
4.1. Formalization

We aim for a single change that alters the output of a classifier for many instances simultaneously – see Fig. 2 for an illustration. For this purpose, we propose a concept called *(almost) set consistent counterfactual explanation* that computes a single change $\vec{\delta}_{cf}$ (called counterfactual) that simultaneously changes the prediction for all (or as many as possible) instances $\vec{x}_i \in \mathbb{R}^d$ in a given set $\mathscr{D}$. Different from to Definition 1, we phrase the computation of such a counterfactual $\vec{\delta}_{cf}$ as a multi-objective optimization problem as stated in Definition 2.

Definition 2. ((Almost) Set Consistent Counterfactual Explanation). *Let* $h : \mathbb{R}^d \to \mathscr{Y}$ *denote a prediction function, and let* $\mathscr{D}$ *be a set of labeled instances with the same prediction* $y \in \mathscr{Y}$ *under* $h(\cdot)$ *– i.e.* $h\left(\vec{x}_i\right) = y \quad \forall \vec{x}_i \in \mathscr{D}$*. We call all pareto-optimal solutions* $\vec{\delta}_{cf} \in \mathbb{R}^d$ *to the following multi-objective optimization problem (almost) set consistent counterfactual explanation:*

$$\underset{\vec{\delta}_{cf} \in \mathbb{R}^d}{min}\left(\theta\left(\vec{\delta}_{cf}\right)\ ,\ \sum_{\vec{x}_i \in \mathscr{D}} \ell\left(h\left(\vec{x}_i + \vec{\delta}_{cf}\right), y_{cf}\right)\right) \tag{2}$$

where $\theta(\cdot)$ denotes the cost of the counterfactual, and $\ell(\cdot)$ denotes a suitable loss function penalizing deviations from the requested outcome y_{cf}. Note that the main difference to a standard counterfactual explanation Eq. (1) from Definition 1 is the sum over all instances in $\mathscr{D}$ – i.e. the change $\vec{\delta}_{cf}$ must be valid for all (or as many as possible) instances in $\mathscr{D}$. We propose to solve Eq. (2) by using an evolutionary method such as NSGA-II [51] – by this, we can deal with specific feature constraints (e.g. binary features) and arbitrary classifiers $h(\cdot)$. However, in some special cases, such as linear classifiers, a more efficient way for solving Eq. (2) is possible – see Appendix A for details. Our proposed



(a) Illustration of standard counterfactual explanations: Each instance $\vec{x}_i$ gets a **different** counterfactual explanation $\vec{\delta}_{i_{cf}}$.

(b) Illustration of a multi-instance counterfactual explanation: Each instance $\vec{x}_i$ gets the **same** counterfactual explanation $\vec{\delta}_{\mathrm{cf}}$.

Fig. 2. Illustration of (a) Standard counterfactuals vs. (b) Multi-instance counterfactual – note that in (b) all counterfactuals δ_{cf} are the same.

modeling (Definition 2), together with its rigorous formalization (see Appendix A), allows efficient computation of multi-instance counterfactual explanations (see case study in Section 6) with formal guarantees in contrast to existing work which is usually heuristic based [10,12]. Furthermore, the modeling does not make any assumptions on the model $h(\cdot)$ or the data domain, and is easily extendable which becomes handy in the case of additional constraints, such as specific feature ranges, plausibility constraints, etc.

4.1.1. *Computing diverse counterfactuals*

In practice, it might be beneficial to collect a set of different explanations – i.e. many different changes $\vec{\delta}_{\mathrm{cf}}$ – that all simultaneously change the outcome of many instances as requested, so that the decision-makers (e.g. business owners) can choose among them. Existing methods for computing multi-instance counterfactuals [33], however, do not support the computation of a set of diverse multi-instance counterfactuals. The flexibility of our modeling of multi-instance counterfactual explanations (see Section 4), however, allows us to extend Definition 2 to support diversity as stated below.

Computing diverse multi-instance counterfactuals can be realized by sampling with high variance from the Pareto-front of Eq. (2) – or at least obtaining high-variance samples that are close to the Pareto-front. To compute multi-instance counterfactuals Eq. (2) that differ completely in their used features – i.e. no counterfactual is allowed to share any recommended changes with other counterfactuals –, we propose an iterative method that computes a set of highly diverse approximate (almost) set consistent counterfactuals. For this, we need a mechanism that allows excluding already used features from future counterfactuals.

Algorithm 1. Computation of Diverse (Almost) Set Consistent Counterfactuals.

Input: Instances $\mathcal{D}$, Requested prediction y_{cf}, $k \geq 1$: number of diverse counterfactuals, classifier $h(\cdot)$
Output: Set of diverse counterfactuals $\mathcal{R} = \{\delta_{\mathrm{cf}_i}\}$

$\mathcal{F} = \{\}$ ▷ Initialize set of black-listed features
$\mathcal{R} = \{\}$ ▷ Initialize set of diverse counterfactuals
for $i = 1, \dots, k$ **do** ▷ Compute k diverse counterfactuals
  $\delta_{\mathrm{cf}_i} = \mathrm{CF}_h(\mathcal{D}, y_{\mathrm{cf}}, \mathcal{F})$ ▷ Compute next counterfactual
  $\mathcal{R} = \mathcal{R} \cup \{\delta_{\mathrm{cf}_i}\}$
  $\mathcal{F} = \mathcal{F} \cup \{j \mid (\delta_{\mathrm{cf}_i})_j \neq 0\}$ ▷ Update set of black-listed features
end for

Because of its flexibility, we can modify our optimization problem Eq. (2) not to use any black-listed features $\mathscr{F}$ – i.e. features already used by previous counterfactuals – by introducing a diagonal matrix $\mathbf{M} \in \mathbb{R}^{d \times d}$ and replacing all occurrences of $\vec{\delta}_{\mathrm{cf}}$ by $\mathbf{M}\vec{\delta}_{\mathrm{cf}}$ with $(\mathbf{M})_{i,i} = 0$ if $i \in \mathscr{F}$, 1 otherwise. The effect of this is that it sets all changes in black-listed features $\mathscr{F}$ to zero – i.e. it removes forbidden changes and the solver is required to find other changes that yield a feasible solution. Note that this extension does not change the complexity of the previous optimization problems Eq. (2). However, there are cases where for a set of black-listed features $\mathscr{F}$ no feasible solution exists – i.e. no allowed changes will yield the requested outcome.

We introduce the following notation: $\mathrm{CF}_h\left(\mathscr{D}, y_{\mathrm{cf}}, \mathscr{F}\right)$ computes the solution to one of the previous optimization problems (i.e. depending on $h(\cdot)$ Eq. (2) under the additional constraint that no black-listed features $\mathscr{F}$ must be used in the final solution. The complete procedure is described in pseudo-code in Algorithm 1.

4.2. *Comparison of the proposed multi-instance counterfactual explanations method with other techniques*

The realization of our multi-instance counterfactual explanations differs from existing methods in the literature since it does not make any assumption on the classifier $h(\cdot)$, nor on the data domain, and thus can be applied to different scenarios. To the best of our knowledge, the only other method with the same properties is the one proposed in [10]. Also, in contrast to other methods the proposed method's implementation is derived from a rigorous mathematical formalization (see Appendix A) which enables us to state formal guarantees on the results as well as to develop efficient scenario-specific algorithms. Finally, the proposed method can be extended and customized to compute diverse multi-instance counterfactuals (see Algorithm 1) in contrast to other methods in the literature.

5. Improving customer repurchase intention using multi-instance counterfactual explanations

The proposed methodology for generating optimum and low-cost recommendations for businesses on how to improve customer repurchase, combines topic modeling with multi-instance counterfactual explanations.

Customers usually express their opinions through eWOM, thus topics that emerge from topic modeling are used as a proxy to different servicescape factors, the performance of which shape customers' experience and repurchase behavior. In counterfactual terms, we are seeking to find optimum business changes that will alter discussed topics in eWOM – i.e. topics that are expected to be (or not) discussed in reviews to observe customer repurchase behavior (e.g. revisit in the case of a restaurant). Therefore, to inflict the desired changes to eWOM's topics, organizations need to address the poorly performing factors (e.g. servicescape) that caused topics to be mentioned in the first place. The recommendations from our proposed counterfactual explanation method refer to factors the business needs to improve, to increase repurchase intentions for multiple customers.

The proposed methodology uses clustering to group customers with similar needs/opinions, and use the multi-instance counterfactual explanation method to address the needs of each group separately. By doing this, we provide a more detailed understanding of the customers' opinions and allow the business to target customer groups that it considers more important(e.g. profitable). To provide decision-makers with more options and increase their confidence in the changes being recommended, multiple diverse counterfactuals (i.e. recommendations) are generated.

The main steps of our proposed methodology are the following:

1. *Data preparation:* Collect eWOM from social media platforms (e.g. Yell, TripAdvisor, etc.) regarding customers of the business/organization under study. Extract eWOM's textual information and customer ratings. To examine the problem of repurchase behavior, sufficient reviews must be collected to identify customers with a sufficient number of repeated purchases. From revisit cases, we keep the first review (chronological) from all revisits of the same customer to the same venue. Customers with a low or no repurchase intention are identified based on negative reviews (e.g. less than 3-star rating) and only one visit(purchase) to the same venue. The above heuristics are used to build a binary labeled dataset with positive reviews referring to customers with a high repurchase intention and negative reviews to customers with a low repurchase intention.
2. *Opinion mining:* Build a topic model using the textual part of the filtered reviews and use the topics distribution per review (proportion of each topic in the review) as reviews' embeddings (vectorization of reviews).
3. *Binarization of reviews:* Topic embeddings from the previous step are binarized based on a user-specified threshold, thus topics probabilities less or greater than the threshold are assigned to zero/one accordingly. Binarization is performed since categorical features are easier to understand by people according to [52] and thus, improves comprehension of the recommended business changes.
4. *Predictive modeling:* Train a binary classifier to predict a customers' repurchase intention using the binarized topic embeddings of reviews.
5. *Recommendations for groups of customers:* Use the embeddings of reviews in the test-set, that are correctly classified as low repurchase intention, to cluster customers into groups with similar needs – This can be done either manually or using a clustering technique such as K-modes.

For each group of customers compute one or multiple diverse multi-instance counterfactual explanations. These refer to changes that need to be made to the binarized topics to change the prediction of the model from "low repurchase intention" to "high repurchase intention" for all (or many) customers in the same group. Changes in topics refer to modifications that the organization needs to implement(i.e. with regards to servicescape) so that customers do or do not discuss these in eWOM.

We interpret the computed (group-wise) recommendations in the last step as changes that the business needs to implement to increase repurchase intentions for each group of customers.

6. Case study

We apply the proposed methodology (see Section 5) to a case study from the hospitality industry, using real data from TripAdvisor. The focus is on recommending optimum changes that restaurants at a touristic destination need to make, with regards to their servicescape (see Section 2), to improve revisit performance. Our approach for identifying servicescape factors is similar to [25], which analyses eWOM through topic modeling and classifies reviews according to the five dimensions of the service quality evaluation model (SERVQUAL). However, unlike the work in [25], our method focuses on identifying optimum & actionable & low-cost changes to servicescape to improve revisit performance through our proposed multi-instance counterfactual explanation method.

6.1. Data

The utilized data refers to 250 k reviews (English language) from customers who visited restaurants in Cyprus between the years 2010 to 2020 and posted their opinions (publicly available) about their experience on TripAdvisor. The total number of unique customers is 123 k and the number of restaurants is 2615.

Our proposed methodology is generic and thus can be applied to different business scenarios such as, a touristic destination (city or a region). In such a scenario the destination's management organization can utilize the method to identify optimum and actionable changes to their policies and/or servicescape – for instance, minimum level of employee training, minimum salary to keep them motivated, minimum quality of food ingredients, cleanliness etc. – that will improve destination's revisitation by leveraging on factors that cause revisits. Alternatively, the method can be applied to data of a specific restaurant or a chain of restaurants, to identify optimum changes to servicescape that will improve revisits. In each scenario, eWOM data need to be filtered to include only reviews relevant to the organization/destination being investigated.

Our case study is based on the first scenario and the destination under scrutiny is Limassol, a coastal city in the Republic of Cyprus. Therefore, the initial eWOM data is filtered to include only reviews from this destination. The filtering reduces the initial data set of 250 k to 27 k reviews, 8 k of which refer to re-visitors, from 21 k unique users. Data labeling is performed by identifying revisitors (high repurchase intention) and non-revisitors (low repurchase intention) from the initial dataset. Customers who visited the same venues more than once and whose evaluation score is greater than $\frac{4}{5}$ are labeled as revisitors, while customers who visited a restaurant only once and had a review rating less than $\frac{3}{5}$ (negative review) are labeled as non-revisitors. All other reviews from the initial dataset are ignored. Furthermore, the textual reviews underwent cleaning through punctuation and URLs elimination, lowering of text, stop-words removal, tokenization, and text normalization through contractual expansion of abbreviations (i.e., "do not" to "don't"). Domain-specific stop-words were identified by considering irrelevant tokens in the topics of the learned topic model and refining the model by adding irrelevant words to the list of common stop-words. Thus, names of people, restaurants, cities, foods, etc. were gradually added as extra stop-words.

6.2. Setup

The proposed methodology (see Section 5) is applied to the aforementioned restaurant revisit problem through the following steps:

1. *Data preparation:* We label reviews collected from TripAdvisor for Limassol-Cyprus.
2. LDA is used in the topic modeling step (i.e. extracting customers' opinions). Prepossessed reviews are also filtered using a part of speech tagger, to include only: nouns, adjectives, verbs, adpositions, interjections, and adverbs. The optimum number of topics, based on semantic coherence score was 20, as depicted in Fig. 3. The interpretation and naming of the topics is based on domain knowledge and the most prevalent words associated with each topic. To model only highly discussed issues, the topics were binarized based on a threshold value of 0.5.

   Customers' opinions (modeled as topics) in Fig. 3 reflect on different dimensions of restaurants' servicescape, with factors such as: the quality of the service, cleanliness, price, complaint management, food quality, location, music/entertainment, etc., being highly related to revisiting behavior, as reported in the literature [13,17].

   Fine-tuning of the analysis can be performed by selecting the topics to be manipulated a priori. This can be done by focusing on topics that can be easily implemented by the management (actionability) or on factors on which the analyst wants to focus the investigation. In that regard, we manually selected the following 16 topics to be manipulated during the analysis: *"reservation issues", "place with music", "dry cold food", "dispute with seating and ordering", "bad taste", "local cuisine", "toilet not clean old place", "family meal", "menu choice/set menu", "poor service but good location and view", "long wait for food", "rude staff", "staff friendliness", "three course meal", "price and portions", "food freshness"*.
3. *Predictive modeling:* Binarized topics per review (i.e. review embedding) are used as features during training of an XGBoost binary classifier that predicts revisit intention. Model training & testing is performed using 5-fold cross-validation. The average F1-score of the trained classifier is approximately 82%.
4. *Recommendations for groups of customers:* Binarized embeddings of reviews are used to cluster non-revisiting customers. Given categorical nature of the the data (binary), we thus use k-modes to cluster non-revisiting customers into 5 clusters. The optimum number of clusters was determined automatically by optimizing the within-cluster sum of squares. For each cluster of non-revisiting customers, the proposed multi-instance counterfactual explanations (see Section 4) is applied along with each trained classifier from each of the 5-folds, to generate consistent and cost-effective recommendations that satisfy all (or most) non-revisiting customers of each cluster.

We evaluate the generated recommendations quantitatively as well as qualitatively. On a quantitative level, we evaluate the correctness (see Table 1a) of the recommendations based on the predictions of the revisit intention classifier – i.e. we evaluate for how many of the non-revisit instances, the recommendations change the output of the revisit intention classifier. The cost and complexity of the recommendation are presented in Table 1b. Cost is defined as the number of factors (topics) that have to be changed. We evaluate the effect of the clustering on the computed recommendations by comparing the computed recommendations per cluster with the recommendation computed for a single (global) cluster – i.e. all non-revisitors are assigned to the same cluster. Furthermore, we evaluate the runtime (see Table 2) – i.e. comparing how much time it takes to compute the recommendations in each case. The multi-instance counterfactual explanations are computed using our proposed method from Section 4. The results of our method are compared against the method by Warren et al. [10], since it is the only method applicable in this scenario (i.e. pre-defined groups of customers that are represented by binary data). Because our proposed method also

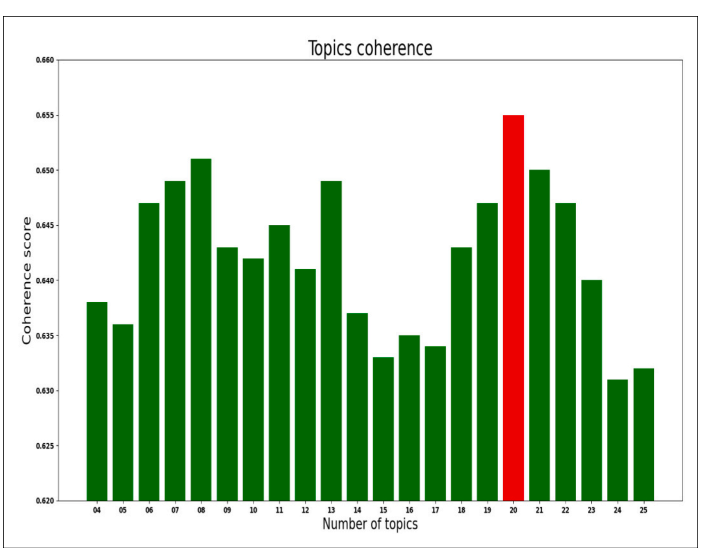
Table 1
Correctness and complexity scores

(a) Correctness of the computed recommendations. We report the mean and variance (over all folds) rounded to two decimal points – larger numbers are better

Method	No Clustering	Clustering
Ours	1.0 ± 0	1.0 ± 0
Warren [10]	0.34 ± 0	0.90 ± 0.04

(b) Complexity (i.e. number of recommended changes) of the computed recommendations. We report the mean and variance (over all folds) rounded to two decimal points – smaller numbers are better.

Method	No Clustering	Clustering
Ours	9.9 ± 6.6	7.8 ± 3.36
Warren [10]	15 ± 0	15 ± 0



Topic Label	Top words per topic based on probability
Reservation issues	table,book,day,night,reservation,dinner
Place with music	place,music,visit,area,time,open,food
Dry cold food	order,dry,cold,cook,serve,look,food
Dispute with seating and ordering	table,ask,sit,look,waiter,come,order,leave,
Bad taste	bad,taste,place,eat,really,terrible,food
Local cuisine	cypriot,place,local,dish,tourist,good
Regular customer visiting again	visit,year,time,service,staff,make,good
Toilet not clean old place	make,toilet,clean,kitchen,need,hand
Complains	ask,say,order,tell,waiter,manager,pay,bring
Family meal	meal,eat,cook,say,child,order,husband,sea,
Based on reviews	review,read,good,meal,look,decide,write
Poor service but good location and view	service,poor,good,location,quality,view
Menu choice/ set menu	menu,choice,good,dish,set,choose,price
Long wait for food	order,wait,minute,arrive,drink,come,hour,
Rude staff	staff,rude,customer,service,manager,waiter
Staff friendliness	good,staff,friendly,nice,service,place
Three course meal	main,starter,course,meal,start,dessert
Price and portions size	price,portion,small,quality,place, expensive
Food freshness	taste,fresh,dish,cook, fry,grill,platter
Drinks	drink,bar,beer,water,bottle,glass,cocktail

Fig. 3. Optimum number of topics during topic modeling (left) and the defined labels (naming) of topics based on the distribution of words per topic (right).

Table 2
Time (*sec*) it takes to compute the recommendations. We report the mean and variance (over all folds) rounded to two decimal points – smaller numbers are better.

Method	No Clustering	Clustering
Ours	13.10 ± 5.65	11.24 ± 2.71
Warren [10]	290.53 ± 45.78	56.27 ± 125.25

supports diversity (see Section 4.1.1), we additionally compute up to three diverse counterfactuals for each setting (group of customers) and evaluate those as well (see Table 3) – note that none of the other methods from the literature supports the computation of multi-instance counterfactual explanations.

On a qualitative level, we manually evaluate generated recommendations by checking how well they align with the known relevant factors from the literature. Furthermore, we also compare the recommended changes (i.e. changes to certain factors) with the most relevant factors identified using SHAP [53]. Because the method by Warren et al. [10], often fails to generate correct solutions (see Table 1a), we only consider the recommendations generated by our proposed method.

6.3. Results and discussion

6.3.1. Quantitative evaluation

We observe that in all cases the correctness of recommendations generated by our proposed method is almost always 100% – i.e. the recommended changes indeed change the revisit intention of all individuals in the clusters. As seen in Table 1a, the correctness of the recommendations generated by the method proposed by Warren et al. [10] is significantly worse than the ones computed by our method. However, the correctness increases significantly when just considering a sub-group (i.e. cluster) of non-revisitors. This provides evidence that our proposed clustering step in the methodology makes it easier to compute recommendations on how to increase the revisit intention.

We also note a big difference in the runtime performance of both methods. The method proposed in [10] is more than 20 times slower than our proposed method. Although runtime performance depends on the specification of the different machines used for the analysis, we still consider this as a significant advantage of our method and evidence of the computational efficiency of our proposed realization of multi-instance counterfactuals.

Regarding the complexity of the generated recommendations (i.e. number of changes), we observe that clustering non-revisitors into groups leads to simpler recommendations (i.e. fewer number of changes) than in the case of no clustering. However, in the case of Warren et al. [10], clustering does not seem to have any effect on the complexity of the recommendations, while it improves correctness as stated above.

The diverse recommendations generated by our proposed multi-instance counterfactual method (see Section 4.1.1) also have excellent performance in terms of correctness. Furthermore (see Table 3), the results show that complexity is best (i.e. lowest) when using diversity compared to non-diverse multi-instance counterfactuals. This suggests that by computing multiple recommendations, simpler explanations can also be found. Diverse counterfactuals however have higher variance compared to non-diverse explanations. This demonstrates that our proposed method (see Section 4.1.1) is able to generate diverse recommendations with high accuracy and thus allows the decision maker to choose among several possible recommendations.

Table 3
Mean correctness (higher better) and complexity (number of recommended changes– lower better) over all folds of the computed diverse recommendations.

Metric	No Clustering	Clustering
Correctness ↑	1.0 ± 0	1.0 ± 0
Complexity ↓	8.8 ± 3.35	4.62 ± 12.80

6.3.2. Qualitative evaluation

In this section, we discuss a subset of the low-complexity (e.g. few changes) recommendations generated by the diverse multi-instance counterfactuals, for different clusters of customers, since diverse counterfactuals turned out to be the best (complexity) in the quantitative evaluation (i.e. see Table 3).

Examples of these recommendations are presented below. These are consistent with the servicescape literature [17] [13], and thus provide evidence of the method's validity.

Based on the cluster-based option, the following four diverse recommendations are presented. These refer to different clusters of customers. As already stated, diverse multi-instance counterfactuals produce easier recommendations and thus were selected:

1. *If restaurants improve food quality, service and the waiting time of customers at venues with local cuisine THEN revisit intention will increase for customers of cluster 1*. The recommendation highlights to destination authorities the importance of traditional food quality and excellent service. This cluster refers to customers who pay attention to food taste and quality and enjoy local cuisine. To implement this recommendation, authorities could offer training to staff on local cuisine and food preparation management so that the waiting time is reduced.
2. *If restaurants stop offering low-quality set-menu and the staff start being friendly when customers are seated or are waiting THEN revisit intention increases for all customers of cluster 2.* This recommendation can be implemented through the offering of high-quality set-menu, since it is considered value for money by many customers; staff attitude can be addressed through additional training or by having a standard of minimum staff qualifications. Lastly, seating issues due to reservations or the number of customers arriving simultaneously can be tackled through more sophisticated reservation systems.
3. *If restaurants address the long wait for food and the long time for customers to be seated and improve the quality of the food and staff friendliness THEN revisit intentions for customers belonging to cluster 3 will increase*. This recommendation can be implemented by specifying a minimum number of staff per customer served at a venue since service errors are a result of under-staffing or overloading. Long waits for food can be tackled through a Real-Time Process Management System.
4. *If restaurants improve the quality and price of their three course meals and eliminate impolite staff behavior THEN revisit intention would increase for all customers of cluster 4.* This recommendation can be implemented through policy for minimum qualifications for service staff and regular personnel retraining.

The factors used in the above recommendations are verified by the literature highlighting that psychological motives such as those described (staff friendliness, waiting time [54], complain management) prevail over physical attributes of service delivery when it comes to revisits [14] and thus decisive actions are required to restore customer revisit intention.

The recommendations pinpoint aspects that the destination's management can utilize to improve revisitation. An additional strategy for selecting recommendations and servicescape factors to act upon is to consider the monetary benefits associated with each customer cluster (when this information is available). Thus, a cluster of high-spending customers could be considered more important in terms of profitability than a cluster of low-spending customers even though the latter is a larger cluster.

The results from the method can be improved by assigning weights to features (topics), to define the difficulty of implementing changes to servicescape factors associated with each topic. The weighted sum of all topics associated with each counterfactual can be used as a normalizing factor for selecting recommendations that are easier to implement and would have the highest impact. The specification of the topics' weights can be left to the decision maker who can use techniques such as the Analytic Hierarchy Process [55]. For simplicity, in this case study, we assume that all topics have equal weights.

6.3.3. Comparison with feature importance scores

We use SHAP [53] to compute the importance of the revisit intention classifier's features as shown in Fig. 4. SHAP assigns each feature an importance score referring to its contribution to the revisit intention classification. The y-axis shows the features' names in order of importance and the x-axis the SHAP value (i.e. importance score). The association of each feature with revisit or non-revisit instances is based on the intensity of the color of each instance's feature value (shown as dots on the plot) and the positioning of instances on the x-axis. SHAP values greater than zero are associated with a high revisit intention, thus instances positioned on the positive side of the x-axis relate to high revisit intention and vice versa. We observe that the most important features from SHAP align with factors that are often changed in our computed recommendations using the multi-instance counterfactuals method. This provides additional evidence of the correctness of the generated recommendations. However, in contrast to our proposed methodology, SHAP features' importance scores can not be used to determine the minimal set of features that can change the classifier's output, nor can they state the exact magnitude and value of the required changes.

7. Contributions and implications

From the managerial perspective, this work makes several novel contributions regarding how to improve business practices for increasing repurchase. First, it proposes a novel methodology to automatically recommend actionable changes to improve customers' repurchase intentions. Such recommendations can have a significant impact on organizations' profitability and competitiveness since they are optimized on costs (e.g. number and degree of change required) and provide a way to improve repurchase performance. Secondly, the proposed methodology can be applied to different groups of customers, depending on stakeholders' choice, and thus help improve repurchase intentions for customers that are considered important to the business. This is realized through either segmentation of customers based on their opinions (topics) using different clustering techniques or through clustering using a specific attribute such as the money that customers spend [56] (when available) since this is a popular way of segmenting customers in marketing. Thirdly, the methodology utilizes data that is freely available online such as eWOM, and thus enables the automated extraction of customer opinions from text. Given that customer opinions are known to change over time, the method can recommend in a timely fashion optimum business responses to tackle emerging challenges described in eWOM, since it lends itself to automation. Contrary to our methodology are survey-based approaches that suffer from low response rates and time delays [57] that hinder rapid response to the re-visitation problem. Fourthly, the method is generic and thus can be used by group recommender systems to provide restaurant recommendations to groups of customers, by matching their preferences to businesses that require no changes to their servicescape factors to guarantee their revisit/ satisfaction.

From a theoretical perspective, the proposed methodology offer advantages over local counterfactual explanation methods that use individual instances to draw global insights about a model. This is key in many business problems that require the simultaneous satisfaction of multiple entities through minimal changes. The methodology yields results that satisfy the above, are actionable and easily interpreted by stakeholders, since they are expressed in binary form through the use of a topic threshold value. The method is generic and thus can be applied to different problems that require optimum changes to policy/practice to satisfy multiple entities such as employees; road safety issues; government policy changes etc. Considering the limited previous research on the utilization of eWOM's text for optimizing revisit, the proposed methodology can be applied to data from diverse customer segments (e.g. such as different cultures, personalities, etc).

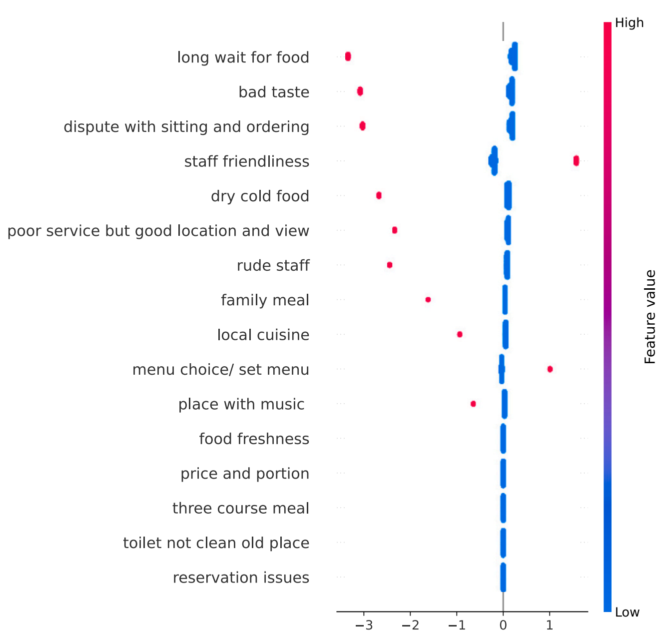


Fig. 4. Global SHAP explanations of revisit intention classifier.

8. Conclusion

The methodology proposed in this work utilizes multi-instance counterfactual explanations to generate recommendations that can be easily implemented by businesses, to satisfy a group of customers. This is key in problems such as customer repurchase, where specific actions need to be implemented by management to satisfy multiple non-repeating customers with low-cost interventions to practices. This method extends previous counterfactual methods that focus on aggregating individual instances (customers) to make recommendations. The proposed methodology is empirically evaluated using real-world data from TripAdvisor. The results highlight that the method is able to come up with recommendations that are actionable and abide with the literature, and thus provide evidence of validity. Recommended actions are expressed in terms of factors extracted from a topic model that refers to prevalent customer opinions in a corpus of customers' reviews. Topics are used by the proposed method as features to compute recommendations that satisfy all or most of the non-revisiting customers to a touristic destination. Future work will focus on enhancing the method, by providing weights to customer opinions to optimize further the generated recommendations, and also utilize causal priors to eliminate implausible and unrealistic recommendations. Moreover, to ease the burden of labeling the topics that emerge from topic modeling, automated topic labeling techniques will be investigated. Finally, to minimize bias from possible fake reviews the approach can be improved by filtering fake eWOM before computing the counterfactuals.

CRediT authorship contribution statement

André Artelt: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Andreas Gregoriades:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Set consistent counterfactuals – a formal approach

In this section, we formalize and investigate the previously discussed intuition & concept of a multi-instance counterfactual: We aim for a small (number of) and cheap changes that, if applied to a set of samples $\mathcal{D} \subset \mathcal{X}^n$, the predictions of a given classifier $h : \mathcal{X} \to \mathcal{Y}$ changes simultaneously to some requested $y \in \mathcal{Y}$. We call such changes *set consistent counterfactuals* and formally define them as follows – note that in [12] the exactly same problem is called *group-wise counterfactual explanation*:

Definition 3. (Set Consistent Counterfactual). *For a prediction function* $h : \mathbb{R}^d \to \mathcal{Y}$ *and a set of instances* $\mathcal{D} = \left\{ \vec{x}_i \right\} \quad \vec{x}_i \in \mathbb{R}^d$ *together with a requested prediction* $y_{cf} \in \mathcal{Y}$ *– whereby* $h\left(\vec{x}_i\right) \neq y_{cf} \quad \forall \vec{x}_i \in \mathcal{D}$ *–, we call a vector of changes* $\vec{\delta}_{cf} \in \mathbb{R}^d$ *a* set consistent counterfactual *if it is a solution to the following optimization problem:*

$$\min_{\vec{\delta}_{cf} \in \mathbb{R}^d} \theta\left(\vec{\delta}_{cf}\right) \quad s.t.\ h\left(\vec{x}_i + \vec{\delta}_{cf}\right) = y_{cf} \quad \forall \vec{x}_i \in \mathcal{D} \tag{A.1}$$

where $\theta((\cdot)$ computes the cost of $\vec{\delta}_{cf}$ – e.g. change as few features as possible. Since a set consistent counterfactual according to Definition 3 might not always exist, we consider a more relaxed version of set consistent counterfactuals as follows (given as Definition 2 in the main text):

Definition 4. (Almost Set Consistent Counterfactual). *For a given classifier* $h : \mathbb{R}^d \to \mathcal{Y}$ *and a set of instances* $\mathcal{D} = \left\{ \vec{x}_i \right\} \quad \vec{x}_i \in \mathbb{R}^d$ *together with a requested prediction* $y_{cf} \in \mathcal{Y}$ *–whereby* $h\left(\vec{x}_i\right) \neq y_{cf} \quad \forall \vec{x}_i \in \mathcal{D}$ *–, we call all pareto-optimal solutions* $\vec{\delta}_{cf} \in \mathbb{R}^d$ *to the following optimization problem an* almost set consistent counterfactual*:*

$$\min_{\vec{\delta}_{cf} \in \mathbb{R}^d} \left(\theta\left(\vec{\delta}_{cf}\right) \quad \sum_{\vec{x}_i \in \mathcal{D}} 1_{\left\{ h\left(\vec{x}_i + \vec{\delta}_{cf}\right) \neq y_{cf} \right\}} \right) \tag{A.2}$$

Next, we state the connection between set consistent counterfactuals (Definition 3) and almost set consistent counterfactuals (Definition 4):

Lemma 1. Any set consistent counterfactual (Definition 3) is an almost set consistent counterfactual (Definition 4).

Proof. A solution $\vec{\delta}_{\mathrm{cf}}$ to Eq. (1) implies that the second objective in Eq. (2) becomes equal to zero which is also its lower bound because it is a sum of indicator functions. By definition, there can not exist some other $\vec{\delta}_{\mathrm{cf}}^{\,*}$ with: $\theta\left(\vec{\delta}_{\mathrm{cf}}^{\,*}\right) < \theta\left(\vec{\delta}_{\mathrm{cf}}\right)$ and $h\left(\vec{x}_i + \vec{\delta}_{\mathrm{cf}}^{\,*}\right) = y_{\mathrm{cf}} \quad \forall \vec{x}_i \in \mathcal{D}$ Therefore, $\vec{\delta}_{\mathrm{cf}}$ is a Pareto-optimal solution to Eq. (2) and Lemma 3 follows. □

We call the counterfactuals according to Definition 4 *almost set consistent counterfactuals* because they might not necessarily be valid for all instances in $\mathcal{D}$ – i.e. for some instances in $\mathcal{D}$, applying $\vec{\delta}_{\mathrm{cf}}$ might not change the outcome as requested. However, this is to be expected in practice – i.e. there might exist a working solution for the majority of samples except for a few instances that might be negligible.

For practical reasons as discussed, we only consider Definition 4 in the remainder of this work, such as our proposed algorithms in Section 4.1.

A.1. Implementation

Linear Classifiers. In the case of a linear binary classifier $h(\vec{x}) = sign(\vec{w}^{\mathrm{T}}\vec{x} + b)$ such as linear-SVM, logistic regression, etc., we can rewrite Eq. (2) into a linear program [39]. Note that linear programs (LP) are special instances of convex optimization problems that can be solved very efficiently. Furthermore, the formalization as a linear program allows us to easily add additional constraints such as plausibility constraints or black-listing some features that must not be changed by the business.

General Classifiers. Without making any assumptions about $h : \mathbb{R}^d \to \mathcal{Y}$, we can utilize an evolutionary method such as NSGA-II [51] for computing almost set consistent counterfactual (Definition 4), as we did in the presented case study in Section 6. The benefit of using an evolutionary method is that it can easily handle additional constraints and any feature domain.

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